

EvoPresent / PresAesth: Self-Improvement Aesthetic Agents for Academic Presentations

Verified reference summary • Year: 2025 • Source bibliography: docs/REFERENCES.md

What the paper contributes

EvoPresent/PresAesth frames academic-presentation design as an iterative self-improvement problem. Instead of relying only on static rules, it uses aesthetic assessment, pairwise preferences, and agent feedback to improve generated presentations over time.

Method / evaluation stance

Its most important contribution for this repo is the calibration mindset: a scoring function is more credible when checked against human preferences or labelled slide pairs. The paper points toward future work where slidekit weights are tuned against a dataset rather than hand-selected.

How slidekit uses it

slidekit already computes deterministic sub-scores, but it does not claim human correlation. This paper supports that restraint. The current YAML approach makes every metric inspectable; future calibration could adjust weights while keeping input structure and formulas transparent.

YAML vs HTML takeaway

HTML/render-first methods are valuable for collecting preference data because humans judge rendered artifacts. YAML/source-first methods are better for applying the learned preferences reproducibly: an agent can fix a field, component, or spacing choice rather than manipulate pixels.

Bottom line

The paper supports a pragmatic split: use research-style rendered artifacts when the question depends on final pixels or human preference collection, but keep slidekit production decks in a constrained YAML IR so layout, linting, and agent fixes remain deterministic, inspectable, and cheap.

Primary slidekit mapping	balance / alignment / whitespace / overlap / contrast / richness / consistency, depending on the paper
Operational risk	HTML metrics can drift with rendering environment; YAML metrics can miss pixel-only effects. Use each where it is strongest.